Akoizo Original Practice - Form 4

Division C | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An observer 8 m from the launcher sees apogee at 45°, with eye height 1.5 m. What is the apogee?

A. 11.3 m

B. 6.5 m

C. 9.5 m

D. 8 m

Answer: _____

2. (2 points) Raising launch pressure mainly increases:

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. The parachute area

B. The nozzle diameter

C. The impulse and launch speed

D. The rocket’s mass

Answer: _____

3. (2 points) Why is changing one variable at a time important in test flights?

A. It shows which change caused the difference in results

B. The rules require it

C. It makes rockets lighter

D. It shortens the log

Answer: _____

4. (2 points) Why do many canopies include a small hole at the top?

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

A. To reduce swinging and steady the descent

B. To make the parachute fall faster for scoring

C. Because the rules require it

D. To attach the ping pong ball

Answer: _____

5. (4 points) Why are cutting, sanding, and gluing the bottle prohibited?

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

Answer: _____
_____
_____

6. (4 points) Explain why hitting the ceiling is usually worse than flying lower.

Answer: _____
_____
_____

7. (4 points) Describe how a passive nose cup releases the payload.

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

Answer: _____
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_____

8. (4 points) Why record peak height in a gym that is lower than the competition venue?

Unit: Apogee, ceiling height, and choosing pressure.
Worked-case data: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?
Separate investigation observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

Answer: _____
_____
_____

9. (4 points) Why is maximum height not the goal?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

Answer: _____
_____
_____

10. (4 points) What should a team do on launch 2 after a strong launch 1?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.

Answer: _____
_____
_____

12. (2 points) Define “Pressure vessel” in the context of Ping Pong Parachute.

Answer: _____
_____
_____

13. (2 points) Define “Time aloft” in the context of Ping Pong Parachute.

Answer: _____
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14. (2 points) Define “Separation” in the context of Ping Pong Parachute.

Answer: _____
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15. (2 points) Define “Tangent method” in the context of Ping Pong Parachute.

Answer: _____
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16. (2 points) Define “Stability margin” in the context of Ping Pong Parachute.

Answer: _____
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17. (2 points) Define “Multiplier” in the context of Ping Pong Parachute.

Answer: _____
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18. (2 points) Define “Burnout” in the context of Ping Pong Parachute.

Answer: _____
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